

5633032

B.Tech. DEGREE EXAMINATION, MAY 2019.

Third Semester

Computer Science and Engineering

DATA STRUCTURES

(2013-14 Onwards Regulation)

Time : Three hours

Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL Questions.

All Questions carry equal marks.

- 1. Write how to create programs.**
- 2. State any two differences between One Dimensional Array and Multidimensional array.**
- 3. Define Queue.**
- 4. Construe any two applications of Linked List.**
- 5. What is a Binary Tree?**
- 6. Characterize the significance of AVL Tree.**
- 7. State the termininology of Graph.**

8. Define Topological Sort.

9. What are Static tree tables?

10. What is External Sorting?

PART B — ($5 \times 11 = 55$ marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Expound the Pointer Arrays.

Or

12. Explain Selection Sort with suitable example.

UNIT II

13. Interpret the various operations on Queue.

Or

14. Describe the Doubly Linked List with its operations.

UNIT-III

15. Explicate the representation of Binary Tree.

Or

16. Stipulate the Lower and upper bounds of a B Tree.

UNIT IV

17. Illustrate the Traversals of a Graph.

Or

18. Explicate the Operations on sets.

UNIT V

19. Explain Rectangular tables with its functions.

Or

20. Elucidate External Sorting with its devices.